THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 120	Legs 20 and 48 since $5^2 + 12^2 = 13^2$
2	**(d)** 972	Sides 27 and 36; area $= 27 \times 36$
3	**(a)** 15, 112, 113	Use $(n, \frac{n^2-1}{2}, \frac{n^2+1}{2})$ for odd n and $(n, \frac{n^2}{4} - 1, \frac{n^2}{4} + 1)$ for even n
4	**(a)** 5280	Side 73; half other diagonal 55; area $= \frac{1}{2} d_1 d_2 = 5280$
5	**(b)** 17	Foot distances $\sqrt{13^2 - 12^2} = 5$ and $\sqrt{13^2 - 5^2} = 12$ add up to the width
6	**(a)** 36	Side $= 6$ cm, so area $= 6^2 = 36$
7	**(c)** 9	Foot distance changes from 16 m to 25 m
8	**(c)** 120	Legs 24 and 45 since $8^2 + 15^2 = 17^2$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$6^2 + 8^2 = 36 + 64 = 100 = 10^2$; R is the converse of the theorem.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$25 + 144 = 169$ and $9 + 16 = 25$, so both are true; but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let legs be $3x$, $4x$; hypotenuse $= 5x = 35$ (1); $x = 7$, legs 21 cm and 28 cm (1).	1+1
12	Triplet $(19, 180, 181)$ (1); check $19^2 + 180^2 = 32761 = 181^2$ (1).	1+1
13	$d^2 = 50^2 - 48^2 = 196$ (1); $d = 14$ m (1).	1+1
14	Let legs be $5x$, $12x$; hypotenuse $= 13x = 65$ (1); $x = 5$, legs 25 cm and 60 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Foot to first wall $= \sqrt{37^2 - 35^2} = 12$ m (1); foot to second wall $= \sqrt{37^2 - 12^2} = 35$ m (1); width $= 47$ m (1).	1+1+1
16	Side $= 73$ cm; half-diagonal $= 48$ cm (1); other half $= \sqrt{73^2 - 48^2} = 55$ cm, other diagonal 110 cm (1); area $= \frac{1}{2} \times 96 \times 110 = 5280$ cm² (1).	1+1+1
17	East distance $= 24$ km, south distance $= 45$ km (1); distance$^2 = 576 + 2025 = 2601$ (1); distance $= 51$ km (1).	1+1+1
	OR: Foot to first wall $= \sqrt{53^2 - 45^2} = 28$ m (1); foot to second wall $= \sqrt{53^2 - 28^2} = 45$ m (1); width $= 73$ m (1).	1+1+1
18	Side $= 85$ cm; half-diagonal $= 13$ cm (1); other half $= \sqrt{85^2 - 13^2} = 84$ cm, other diagonal 168 cm (1); area $= \frac{1}{2} \times 26 \times 168 = 2184$ cm² (1).	1+1+1
	OR: East distance $= 24$ km, south distance $= 70$ km (1); distance$^2 = 576 + 4900 = 5476$ (1); distance $= 74$ km (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\sqrt{13^2 - 5^2} = \sqrt{144} = 12$ m (1). (ii) $(5, 12, 13)$ (1). (iii) $\sqrt{13^2 - 12^2} = \sqrt{25} = 5$ m (1). (iv) $12 - 5 = 7$ m (1+1).	1+1+1+2
20	$AC^2 = 5^2 + 12^2 = 169$, $AC = 13$ cm (1); $AC^2 + CD^2 = 169 + 7056 = 7225 = DA^2$ so $\angle ACD = 90^\circ$ (2); area $= \frac{1}{2} \times 5 \times 12 + \frac{1}{2} \times 13 \times 84 = 576$ cm² (2).	1+2+2
	OR: (i) $\sqrt{65^2 - 56^2} = 33$ m (2); (ii) new height 52 m, $\sqrt{65^2 - 52^2} = 39$ m (2); (iii) $39 - 33 = 6$ m (1).	2+2+1

Mistakes to avoid:

- Applying $c^2 = a^2 + b^2$ with the wrong side as hypotenuse; c is always the side opposite the right angle (the longest).
- Forgetting to take the square root at the end (giving 169 instead of 13).
- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9 + 16} = 5$, not $3 + 4 = 7$.
- Assuming three consecutive numbers such as 2, 3, 4 form a right triangle; always verify $a^2 + b^2 = c^2$ (here $4 + 9 \neq 16$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 2184	Side 85; half other diagonal 84; area $= \frac{1}{2}d_1d_2 = 2184$
2	**(c)** 23	Foot distances $\sqrt{17^2-15^2} = 8$ and $\sqrt{17^2-8^2} = 15$ add up to the width
3	**(c)** 64	Side $= 8$ cm, so area $= 8^2 = 64$
4	**(b)** 4	Foot distance changes from 36 m to 40 m
5	**(c)** 112	Legs 14 and 48 since $7^2 + 24^2 = 25^2$
6	**(d)** 480	Sides 16 and 30; area $= 16 \times 30$
7	**(b)** 22, 120, 122	Use $(n, \frac{n^2-1}{2}, \frac{n^2+1}{2})$ for odd n and $(n, \frac{n^2}{4}-1, \frac{n^2}{4}+1)$ for even n
8	**(c)** 5544	Side 85; half other diagonal 77; area $= \frac{1}{2}d_1d_2 = 5544$
9	**(c)** A is true but R is false.	The hypotenuse is opposite the largest angle, so A is true; R is false since it holds only for the hypotenuse.
10	**(d)** A is false but R is true.	Diagonal $= 5\sqrt{2} \approx 7.07$ cm, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Triplet $(24, 143, 145)$ (1); check $24^2 + 143^2 = 21025 = 145^2$ (1).	1+1
12	$d^2 = 65^2 - 63^2 = 256$ (1); $d = 16$ m (1).	1+1
13	Let legs be $8x$, $15x$; hypotenuse $= 17x = 68$ (1); $x = 4$, legs 32 cm and 60 cm (1).	1+1
14	Triplet $(33, 544, 545)$ (1); check $33^2 + 544^2 = 297025 = 545^2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Foot to first wall $= \sqrt{65^2-63^2} = 16$ m (1); foot to second wall $= \sqrt{65^2-33^2} = 56$ m (1); width $= 72$ m (1).	1+1+1
16	Side $= 85$ cm; half-diagonal $= 36$ cm (1); other half $= \sqrt{85^2-36^2} = 77$ cm, other diagonal 154 cm (1); area $= \frac{1}{2} \times 72 \times 154 = 5544$ cm² (1).	1+1+1
17	East distance $= 80$ km, south distance $= 84$ km (1); distance$^2 = 6400 + 7056 = 13456$ (1); distance $= 116$ km (1).	1+1+1
	OR: Foot to first wall $= \sqrt{61^2-60^2} = 11$ m (1); foot to second wall $= \sqrt{61^2-11^2} = 60$ m (1); width $= 71$ m (1).	1+1+1
18	Side $= 89$ cm; half-diagonal $= 39$ cm (1); other half $= \sqrt{89^2-39^2} = 80$ cm, other diagonal 160 cm (1); area $= \frac{1}{2} \times 78 \times 160 = 6240$ cm² (1).	1+1+1
	OR: East distance $= 18$ km, south distance $= 80$ km (1); distance$^2 = 324 + 6400 = 6724$ (1); distance $= 82$ km (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\sqrt{40^2+30^2} = 50$ m (1). (ii) $40 + 30 = 70$ m (1). (iii) $70 - 50 = 20$ m (1). (iv) $15^2 + 20^2 = 225 + 400 = 625 = 25^2$, so it is a triplet (1+1).	1+1+1+2
20	$AC^2 = 12^2 + 16^2 = 400$, $AC = 20$ cm (1); $AC^2 + CD^2 = 400 + 441 = 841 = DA^2$ so $\angle ACD = 90^\circ$ (2); area $= \frac{1}{2} \times 12 \times 16 + \frac{1}{2} \times 20 \times 21 = 306$ cm² (2).	1+2+2
	OR: (i) $\sqrt{65^2-60^2} = 25$ m (2); (ii) new height 52 m, $\sqrt{65^2-52^2} = 39$ m (2); (iii) $39 - 25 = 14$ m (1).	2+2+1

Mistakes to avoid:

- Applying $c^2 = a^2 + b^2$ with the wrong side as hypotenuse; c is always the side opposite the right angle (the longest).
- Forgetting to take the square root at the end (giving 169 instead of 13).
- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9+16} = 5$, not $3 + 4 = 7$.
- Assuming three consecutive numbers such as 2, 3, 4 form a right triangle; always verify $a^2 + b^2 = c^2$ (here $4 + 9 \neq 16$).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 11	Foot distance changes from 40 m to 51 m
2	**(a)** 180	Legs 18 and 80 since $9^2 + 40^2 = 41^2$
3	**(d)** 420	Sides 20 and 21; area $= 20 \times 21$
4	**(a)** 26, 168, 170	Use $(n, \frac{n^2-1}{2}, \frac{n^2+1}{2})$ for odd n and $(n, \frac{n^2}{4} - 1, \frac{n^2}{4} + 1)$ for even n
5	**(d)** 6240	Side 89; half other diagonal 80; area $= \frac{1}{2} d_1 d_2 = 6240$
6	**(c)** 49	Foot distances $\sqrt{41^2 - 40^2} = 9$ and $\sqrt{41^2 - 9^2} = 40$ add up to the width
7	**(c)** 144	Side $= 12$ cm, so area $= 12^2 = 144$
8	**(b)** 6	Foot distance changes from 33 m to 39 m
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\sqrt{12^2 + 5^2} = \sqrt{169} = 13$, obtained by applying the theorem in one of the right triangles.
10	**(c)** A is true but R is false.	$4^2 + 5^2 = 41 \neq 36 = 6^2$ so A is true; R is false since it holds only in right-angled triangles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d^2 = 85^2 - 77^2 = 1296$ (1); $d = 36$ m (1).	1+1
12	Let legs be $7x$, $24x$; hypotenuse $= 25x = 75$ (1); $x = 3$, legs 21 cm and 72 cm (1).	1+1
13	Triplet $(36, 323, 325)$ (1); check $36^2 + 323^2 = 105625 = 325^2$ (1).	1+1
14	$d^2 = 89^2 - 80^2 = 1521$ (1); $d = 39$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Foot to first wall $= \sqrt{85^2 - 84^2} = 13$ m (1); foot to second wall $= \sqrt{85^2 - 40^2} = 75$ m (1); width $= 88$ m (1).	1+1+1
16	Side $= 97$ cm; half-diagonal $= 65$ cm (1); other half $= \sqrt{97^2 - 65^2} = 72$ cm, other diagonal 144 cm (1); area $= \frac{1}{2} \times 130 \times 144 = 9360$ cm² (1).	1+1+1
17	East distance $= 80$ km, south distance $= 150$ km (1); distance$^2 = 6400 + 22500 = 28900$ (1); distance $= 170$ km (1).	1+1+1
	OR: Foot to first wall $= \sqrt{73^2 - 55^2} = 48$ m (1); foot to second wall $= \sqrt{73^2 - 48^2} = 55$ m (1); width $= 103$ m (1).	1+1+1
18	Side $= 109$ cm; half-diagonal $= 60$ cm (1); other half $= \sqrt{109^2 - 60^2} = 91$ cm, other diagonal 182 cm (1); area $= \frac{1}{2} \times 120 \times 182 = 10920$ cm² (1).	1+1+1
	OR: East distance $= 72$ km, south distance $= 21$ km (1); distance$^2 = 5184 + 441 = 5625$ (1); distance $= 75$ km (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $9^2 + 12^2 = 225 = 15^2$, yes (1). (ii) $7^2 + 24^2 = 625 = 25^2$, yes (1). (iii) $8^2 + 10^2 = 164 \neq 144 = 12^2$, no (1). (iv) Hypotenuse $= \sqrt{400 + 441} = \sqrt{841} = 29$ cm (1); perimeter $= 20 + 21 + 29 = 70$ cm (1).	1+1+1+2
20	$AC^2 = 7^2 + 24^2 = 625$, $AC = 25$ cm (1); $AC^2 + CD^2 = 625 + 3600 = 4225 = DA^2$ so $\angle ACD = 90^\circ$ (2); area $= \frac{1}{2} \times 7 \times 24 + \frac{1}{2} \times 25 \times 60 = 834$ cm² (2).	1+2+2
	OR: (i) $\sqrt{25^2 - 24^2} = 7$ m (2); (ii) new height 20 m, $\sqrt{25^2 - 20^2} = 15$ m (2); (iii) $15 - 7 = 8$ m (1).	2+2+1

Mistakes to avoid:

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